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Anthropometric-based predictive equations developed with multi-component models for estimating body composition in athletes. European Journal of Applied Physiology, Ahead of Print: 1-16, 2024.

Association between magnetic resonance imaging use and recovery time following concussion: A CARE Consortium study. Research in Sports Medicine, Ahead of Print: 1-13, 2025.

Athlete selective androgen receptor modulators abuse: A systematic review. The American Journal of Sports Medicine, Ahead of Print: 1-11, 2025.

Coronary microvascular dysfunction years after cessation of anabolic androgenic steroid use. JAMA Open, 7(12): e2451013, 2024.

Current evidence for the use of jugular vein compression collars in sport: A systematic review. Current Sports Medicine Reports, 24(1): 18-27, 2025.

Decreased isometric neck strength is a risk factor for head, neck and face injuries in professional rugby league players. Journal of Sports Sciences, Ahead of Print: 1-8, 2024.

Does caffeine supplementation affect sleep in athletes? A systematic review of nine randomized controlled trials. Clinical Nutrition ESPEN, 65: 76-85, 2025.

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Effect of cold-water immersion treatment on recovery from exercise-induced muscle damage in the hamstring. European Journal of Sport Science, Online Nov. 27: 1-9, 2024.

Effects of thermal interventions on skeletal muscle adaptations and regeneration: Perspectives on epigenetics: A narrative review. European Journal of Applied Physiology, Ahead of Print: 1-25, 2024.

Factors associated with persisting post-concussion symptoms among collegiate athletes and military cadets: Findings from the NCAA-DoD CARE Consortium. Sports Medicine, Ahead of Print: 1-13, 2025.

Football (soccer) match-derived hamstring muscles residual fatigue can be monitored using early rate of torque development. European Journal of Applied Physiology, Ahead of Print: 1-13, 2024.

Fractional tackles: Leveraging player tracking data for within-play tackling evaluation in American football. Scientific Reports, 15: 2148, 2025.

From food supplements to functional foods: Emerging perspectives on post-exercise recovery nutrition. Nutrients, 16: 4081, 2024.

High rate of return to sport for athletes undergoing articular cartilage restoration procedures for the knee: A systematic review of contemporary studies. The American Journal of Sports Medicine, Ahead of Print: 1-12, 2025.

Histological definition of skeletal muscle injury: A guide to nomenclature along the connective tissue sheath/structure. Sports Medicine, Ahead of Print: 1-5, 2024.

Knee ligament injuries in the National Football League: Impact of field surface type during the 2020-2-2023 season. Cureus, 16(12): e76425, 2024.

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Malpractice liability exposure and the sports medicine team physician: Caring for professional athletes in the National Football League, Major League Baseball, and National Hockey League. The American Journal of Sports Medicine, Ahead of Print: 1-8, 2025.

Nutrition for optimizing immune function and recovery from injury in sports. Clinical Nutrition ESPEN, Ahead of Print: 1-37, 2025.

Orthopedic patients are highly amenable to cannabis-based products for the treatment of musculoskeletal pain. Orthopedics, Ahead of Print: 1-8, 2025.

Performance effects of functional knee brace removal and prolonged use in healthy male athlete: Lower extremity power, acceleration, speed, and agility. Journal of Science and Medicine in Sport, Ahead of Print: 1-6, 2025.

Predicting noncontact injuries of professional football players using machine learning. PLOS ONE, 20(1): e0315481, 2025.

Pseudo-sciatica pain due to a biceps femoris muscle tear. Cureus, 16(11): e74286, 2024.

Relationship between serum Vitamin D levels and chronic musculoskeletal pain in adults: A systematic review. Nutrients, 16: 4061, 2024.

Return to sport after distal biceps tendon repair: A systematic review. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2025.

Role of hip internal rotation range and foot progression angle for preventing Jones fracture during crossover cutting. Journal of Applied Biomechanics, Ahead of Print: 1-7, 2024.

Sleep disturbances and risk of sports injury among collegiate student-athletes. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

Sports Medicine: Nutritional sciences and nutritional biochemistry, focusing on nutritional supplements from the FFNFO Framework to contamination. Life, 14: 1392, 2024.

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The effect of alcohol on subsequent sleep in healthy adults: A systematic review and meta-analysis. Sleep Medicine Reviews, 80: 102030, 2025.

The significance of body surface area to mass ratio for thermal responses to a standardized exercise-heat stress test. Medicine and Science in Sports and Exercise, 57(1): 88-93, 2025.

The value of blood lactate and lactate clearance rate in evaluating the prognosis of athletes with heat illness of varying degrees after high-intensity exercise. BMC Sports Science, Medicine and Rehabilitation, 17: 7, 2025.

Ultrasound as a predictor of time-loss injury for the patellar tendon, Achilles tendon and plantar fascia in Division 1 collegiate athletes. British Journal of Sports Medicine, 59: 1-8, 2025.

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A tackler correctly adhering to the tackle instruction in a front-on, one-on-one torso tackle alters the peak inertial head kinematics of the ball carrier but not the tackler. Journal of Science and Medicine in Sport, Ahead of Print: 1-7, 2024.

Objectives: To evaluate if the tackler correctly adhering, or not, to four different instructions of legal front-on one-on-one torso tackles altered the tackler and/or ball carrier peak inertial head kinematics.

Design: Controlled laboratory study.

Methods: Fifteen rugby-code players measured with three-dimensional optoelectronic motion capture performed two tackle instructions from the Australian National Rugby League coaching manual on under (Dominant National Rugby League) and over (Smother National Rugby League) the ball tackles, and two novel variants of these (under, Dominant, Torso Stick; over, Smother, Pop, Lock). A series of mixed general linear models identified if the tackler adhering (n = 455), or not (n = 139) to the tackle instructions altered peak inertial head kinematics.

Results: The tackler's peak inertial head kinematics did not significantly change whether or not they adhered to each of the tackle instructions. When the tackler did adhere to the instructions, the ball carrier sustained a lower peak inertial head kinematics ($p < 0.01$) in the Smother National Rugby League tackle but higher peak inertial head kinematics in the Smother, Pop, Lock.

Conclusions: The ball carriers' inertial head kinematics but not the tacklers were increased when the tackler adhered to this study's variants of the over and under the ball tackle instructions, suggesting that the tacklers were more effective in their tackle performance than the traditional tackle instructions when adhering to the tackle instruction. Greater adherence to the under the ball instructions suggests that the over the ball instruction is a more challenging technique to learn.

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Alterations in whole muscle quality and physiological cross-sectional area measured with quantitative MRI following ACL injury. Journal of Translational Medicine, 23: 79, 2025.

Background: Emerging evidence suggests that there are morphological and physiological changes to the vastus lateralis after an anterior cruciate ligament (ACL) tear. However, it is unclear whether these alterations are limited to just the vastus lateralis or are more representative of widespread changes across the thigh musculature and/or if these changes precede reconstruction. The purpose of this study was to determine T1ρ relaxation time, a measure of extracellular matrix organization in muscle, and physiological cross-sectional area (PCSA) for muscles of the quadriceps and hamstrings of the ACL-deficient and contralateral limbs soon after ACL injury.

Methods: T1ρ and diffusion tensor magnetic resonance imaging were performed on both limbs of 10 participants after primary ACL tear (< 10 weeks). T1ρ relaxation time and PCSA were calculated for all muscles of the quadriceps and hamstrings. Shapiro-Wilks tests were performed to assess normality. Outcomes were compared between limbs for each muscle of interest with paired t-tests or Wilcoxon signed-rank tests with the alpha level set to 0.05.

Results: T1ρ relaxation times were significantly longer for the vastus lateralis (7.0%), rectus femoris (15.4%), and vastus intermedius (9.4%) of ACL-deficient limb; whereas, relaxation times were similar between limbs for all hamstring muscles. PCSA was smaller for the vastus lateralis (-19.6%), vastus intermedius (-20.9%), vastus medialis (-26.0%), and semitendinosus (-15.0%) of the ACL-deficient limb compared to the contralateral limb.

Conclusions: These results provide evidence that morphological and physiological alterations occur within multiple muscles of quadriceps but not the hamstrings prior to ACL reconstruction. Establishing these differences between the quadriceps and hamstrings suggests there is a differential response within the thigh musculature to an ACL injury, providing a framework for more targeted interventions.

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Anthropometric-based predictive equations developed with multi-component models for estimating body composition in athletes. European Journal of Applied Physiology, Ahead of Print: 1-16, 2024.

Purpose: Body composition can be estimated using anthropometric-based regression models, which are population-specific and should not be used interchangeably. However, the widespread availability of predictive equations in the literature makes selecting the most valid equations challenging. This systematic review compiles anthropometric-based predictive equations for estimating body mass components, focusing on those developed specifically for athletes using multicomponent models (i.e. separation of body mass into ≥ 3 components).

Methods: Twenty-nine studies published between 2000 and 2024 were identified through a systematic search of international electronic databases (PubMed and Scopus). Studies using substandard procedures or developing predictive equations for non-athletic populations were excluded.

Results: A total of 40 equations were identified from the 29 studies. Of these, 36 were applicable to males and 17 to females. Twenty-six equations were developed to estimate fat mass, 10 for fat-free mass, three for appendicular lean soft tissue, and one for skeletal muscle mass. Thirteen equations were designed for mixed athletes, while others focused on specific contexts: soccer (n = 8); handball and rugby (n = 3 each); jockeys, swimming, and Gaelic football (n = 2 each); and futsal, padel, basketball, volleyball, American football, karate, and wheelchair athletes (n = 1 each).

Conclusions: This review presented high-standards anthropometric-based predictive equations for assessing body composition in athletes and encourages the development of new equations for underrepresented sports in the current literature.

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Association between magnetic resonance imaging use and recovery time following concussion: A CARE Consortium study. Research in Sports Medicine, Ahead of Print: 1-13, 2025.

Abstract - Researchers may implement magnetic resonance imaging (MRI) to evaluate neurophysiological metrics (e.g. connectivity) in athletes with sports-related concussion (SRC). MRI usage has been purported to exacerbate symptoms that mimic SRC symptomatology, possibly influencing recovery. The present study examined MRI use on SRC recovery outcomes in collegiate athletes. Ninety student-athletes (45 with MRI acutely post-SRC, 45 non-MRI) from four collegiate sites were analyzed. Multivariable Cox proportional hazard regressions were used to measure the SRC recovery risk [return-to-play (RTP) initiation (RTP-I) and unrestricted RTP] with hazard ratios (HR) and 95% confidence intervals (95% CI). No hazard differences for median days to RTP-I [MRI: 10, interquartile range (IQR) = 7-15; non-MRI: 7 (IQR = 4-9)] were observed (HR = 0.75; 95% CI: 0.28-1.21). No hazard differences for median days to RTP [(MRI: 17 (IQR = 11-60); non-MRI: 12 (IQR = 8-24)] were observed (HR = 0.77; 95% CI: 0.29-1.26). MRI use post-SRC does not affect recovery; researchers/clinicians can employ such a procedure without affecting athlete recovery.

Athlete selective androgen receptor modulators abuse: A systematic review. The American Journal of Sports Medicine, Ahead of Print: 1-11, 2025.

Background: Selective androgen receptor modulators (SARMs) are small-molecule compounds that exert agonist and antagonist effects on androgen receptors in a tissue-specific fashion. Because of their performance-enhancing implications, SARMs are increasingly abused by athletes. To date, SARMs have no Food and Drug Administration approved use, and recent case reports associate the use of SARMs with deleterious effects such as drug-induced liver injury, myocarditis, and tendon rupture.

Purpose: (1) To provide a comprehensive synthesis of the literature pertaining to SARMs from a sports medicine perspective and (2) to provide a better understanding of the clinical effects, treatment protocols, prevalence, and potential contamination associated with athlete-consumed SARMs.

Study design: Systematic review; Level of evidence, 4.

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Methods: A systematic review of the English-language literature from PubMed, Cochrane, and Embase databases was performed according to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Articles relevant to SARM clinical outcomes, elimination profiles, contamination, safety profiles, prevalence, and doping control were included.

Results: A total of 72 articles from 2003 to 2022 were identified for inclusion. The prevalence of SARM use among athletes is estimated to be 1% to 3%. SARM preclinical and clinical studies reported significant increases in lean body mass and side effects-including bone remodeling, testosterone suppression, and kidney, liver, and prostate enlargement. Thirteen case reports described 15 cases of SARM abuse. All described patients were men, with a median age of 32 years (range, 19-52 years), more than half were identified as athletes (8/15), and all ingested SARMs orally for a mean course of 8 weeks. Five patients described in the case reports explicitly denied "illicit drug use," implying patients may believe their use to be legal. Athletes most commonly purchased SARMs online, and most of these compounds have been shown to be contaminated with other substances, contributing to adverse effects. Athletes reported consuming SARMs at much higher doses than clinically studied, which may increase the risk of the reported side effects, such as liver injury, impaired insulin sensitivity, cardiovascular events, and tendon damage.

Conclusion: The results of this systematic review serve to educate sports medicine clinicians and researchers on how to better identify, diagnose, and treat athlete SARM abuse. SARM use is associated with increased muscle mass, hepatotoxicity, cardiotoxicity, tendon damage, and androgenic side effects throughout the body-including prostate enlargement and serum testosterone suppression. Identifying and treating SARM abuse requires taking a thorough substance and supplement use history with open communication, providing literature-supported patient education, negotiating SARM discontinuation, and performing multidisciplinary treatment of adverse events. Athlete SARM abuse is increasingly widespread and unsafe, and public health oversight bodies should advocate for regulation of these gray-market compounds.

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Coronary microvascular dysfunction years after cessation of anabolic androgenic steroid use. JAMA Open, 7(12): e2451013, 2024.

Importance: Long-term use of anabolic androgenic steroids (AASs) is associated with a high risk of left ventricular hypertrophy, heart failure with reduced systolic function, and early sudden death, with the mechanism of progression being understudied. Early and persistent impaired myocardial microcirculation could be of clinical importance and a potential underlying mechanism of frequent and early cardiac disease among individuals with AAS use and a future potential target for intervention.

Objective: To investigate coronary microcirculation by measuring myocardial flow reserve (MFR) in men with current and former AAS use compared with controls with no prior AAS use, using cardiac rubidium 82 (82Rb) positron emission tomography/computed tomography (PET/CT).

Design, setting, and participants: This cross-sectional study included men involved in recreational strength training without established cardiovascular disease grouped according to their history of AAS use. The study was conducted November 24, 2021, to August 16, 2023.

Exposure: Cardiac PET/CT with 82Rb was performed at rest and after adenosine-induced stress.

Main outcome and measure: The primary outcome of this study was the MFR among the study groups; a secondary outcome was the coronary calcium score. By definition, impaired myocardial microcirculation was determined using a cutoff of MFR less than 2, and subclinically impaired microcirculation was determined using a cutoff of MFR less than 2.5.

Results: A total of 90 men (32 with current AAS use, 31 with former AAS use, and 27 controls) were included. Mean (SD) age was 35.1 (8.7) years. Elapsed duration since AAS cessation was a geometric mean of 1.5 (95% CI, 0.9-2.5) years. Eighteen men (58.1%) with former use discontinued AAS use more than 1 year before enrollment. Impaired MFR was present in those with current (6 [18.8%]) and former (1 [3.2%]) use, whereas no impairment was observed among the controls (P = .02). Subclinically impaired MFR was higher among men with current (9 [28.1%]) and former (8 [25.8%]) AAS use than the controls (1 [3.7%]) (P = .02). In a multivariable logistic regression model among men with former AAS use, every doubling of the accumulated weekly duration of AAS use (log2) was independently associated with a factor 2 increase in the risk of impaired MFR less than 2.5 (odds ratio, 2.1; 95% CI, 1.03-4.35; P = .04).

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Conclusions and relevance: In this study, men with former AAS use displayed impaired MFR years after AAS cessation, suggesting impaired coronary microcirculation persists beyond active use.

Current evidence for the use of jugular vein compression collars in sport: A systematic review. Current Sports Medicine Reports, 24(1): 18-27, 2025.

Abstract - Sport-related concussions are a common type of brain injury, and the best treatment is prevention. Recently, external jugular vein compression collars have been worn by National Football League players, but the current evidence is limited. To the best of the authors' knowledge, this is the first comprehensive, up-to-date systematic review addressing the use of jugular vein compression collars for decreasing concussion incidence in high-impact sports and activities. Overall, when compared to control, the results indicated that jugular vein compression collars resulted in significantly reduced white matter alteration, improved short-term neurocognitive outcomes, increased internal jugular vein cross-sectional area, decreased internal carotid artery cross-sectional area, and moderating effects on peak pulse pressure. There was no significant difference in concussion incidence rate between groups. While promising, these findings warrant future research to assess jugular vein compression collar's role in concussion prevention, brain injury, and long-term neurocognitive outcomes. The authors are hopeful for the role jugular vein compression collars will play in the future.

Decreased isometric neck strength is a risk factor for head, neck and face injuries in professional rugby league players. Journal of Sports Sciences, Ahead of Print: 1-8, 2024.

Abstract - Head, neck and face injuries are a concern in contact sports. This exploratory study aimed to establish 1) injury risk factors for head, neck and face injuries and 2) clinical cut-off values related to strength, endurance and proprioception of the cervical spine in a team of professional rugby league players. Pre-season assessments of isometric strength of the flexor, extensor and lateral flexor muscles, endurance of the flexor muscles and joint position error were conducted. Injuries resulting in games missed were recorded. Cross-tabulations were used to determine the unadjusted odds ratios for the

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measures as risk factors for playing season injuries. The unadjusted odds ratio (OR) values indicated that if a player had weaker extensors of the neck (<36.4 kg, p = 0.014; <3.4N/kg, p = 0.014) or asymmetry of isometric strength of their lateral flexor muscles (left-to-right ratio <0.91, p = 0.005), their odds of games missed due to season head, neck and face injuries was increased (OR extensors = 8; 95% CI = 1.5-42.5 OR asymmetry of lateral flexor muscles OR = 12.6; 95% CI = 2.0-79.4). As muscle strength is modifiable, the clinical application of this study would involve targeting players in the team beneath the clinical cut-off value.

Does caffeine supplementation affect sleep in athletes? A systematic review of nine randomized controlled trials. Clinical Nutrition ESPEN, 65: 76-85, 2025.

Objective: This systematic review aimed to assess the effects of caffeine supplements on sleep parameters among professional athletes.

Methods: A systematic search of randomized controlled trials (PROSPERO: CRD42024505377) was performed from 1980 to December 2023 through Web of Science (ISI), Cinahl, Embase, CENTRAL, PubMed/MEDLINE, Scopus, and Scienceopen. The effect of caffeine supplement on all sleep parameters (e.g. duration, quality, insomnia), assessed through objective and subjective methods, was investigated among the athletic community.

Results: Of 1469 records, nine trials were eligible for the current review. The studies showed varying results concerning sleep quality, quantity, efficiency, number of awakenings, sleep onset latency, and other sleep-related variables. These differences in findings may be attributable to factors such as the timing of caffeine consumption in relation to sleep time and the time of exercise, habitual caffeine use, and the dose of caffeine prescribed. Given the nature of caffeine, insomnia following ingestion is likely to occur.

Conclusions: This review explores the mechanisms by which caffeine influences sleep in athletes. While caffeine supplementation may enhance athletic performance, it could have a detrimental effect on sleep and therefore recovery. It is important that supplementation considers individual responses to caffeine so that it does not adversely affect sleep in this population.

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Effect of cold-water immersion treatment on recovery from exercise-induced muscle damage in the hamstring. European Journal of Sport Science, Online Nov. 27: 1-9, 2024.

Abstract - This study investigated the effect of five consecutive days of cold-water immersion (CWI) on recovery from exercise-induced muscle damage (EIMD) in the hamstrings following maximal eccentric contraction (EC) exercise. Eighteen healthy adult women were randomly assigned to a CWI group and a control group (CG) (n = 9/group). Participants performed 10 sets of 10 repetitions of isokinetic EC at 30°/second and underwent maximum voluntary isometric contraction (MVC), delayed onset muscle soreness (DOMS) assessment, straight leg raise (SLR) test, and plasma myoglobin (Mb) measurement. The CWI group received one 14-min session of CWI treatment (14°C) at 1, 25, 49, 73, and 97 h after the EC test, whereas the CG rested in a seated position at the same five time points without receiving treatment. (1) All the dependent variables in the CWI group and CG exhibited significant changes after the EC test ($p < 0.05$). (2) The recovery effect in the CWI group was significantly greater than in the CG in terms of the MVC, DOMS, SLR, and plasma Mb concentration results. MVC increased by 89.3 ± 2.0% on the fourth day ($p < 0.013$), DOMS decreased by 15.4 ± 1.5 mm on the second day ($p < 0.000$), SLR increased by 86.3 ± 1.1% on the second day ($p < 0.014$), and plasma Mb decreased by 436.3 ± 60.8% on the third day ($p < 0.014$). The study indicates that five consecutive days of CWI at 14°C significantly enhance recovery from exercise-induced muscle damage in the hamstrings.

Highlights

- CWI treatment was beneficial for recovery from EIMD after high-intensity exercise.
- All the participants exhibited similar and significant responses in their MVC, DOMS, SLR, and plasma Mb concentration results after the EC test.
- The CWI group demonstrated mostly superior recovery responses compared with the CG after receiving the CWI treatment on 5 consecutive days after the EC test.

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Effects of thermal interventions on skeletal muscle adaptations and regeneration: Perspectives on epigenetics: A narrative review. European Journal of Applied Physiology, Ahead of Print: 1-25, 2024.

Abstract - Recovery methods, such as thermal interventions, have been developed to promote optimal recovery and maximize long-term training adaptations. However, the beneficial effects of these recovery strategies remain a source of controversy. This narrative review aims to provide a detailed understanding of how cold and heat interventions impact long-term training adaptations. Emphasis is placed on skeletal muscle adaptations, particularly the involvement of signaling pathways regulating protein turnover, ribosome and mitochondrial biogenesis, as well as the critical role of satellite cells in promoting myofiber regeneration following atrophy. The current literature suggests that cold interventions can blunt molecular adaptations (e.g., protein synthesis and satellite cell activation) and oxi-inflammatory responses after resistance exercise, resulting in diminished exercise-induced hypertrophy and lower gains in isometric strength during training protocols. Conversely, heat interventions appear promising for mitigating skeletal muscle degradation during immobilization and atrophy. Indeed, heat treatments (e.g., passive interventions such as sauna-bathing or diathermy) can enhance protein turnover and improve the maintenance of muscle mass in atrophic conditions, although their effects on uninjured skeletal muscles in both humans and rodents remain controversial. Nonetheless, heat treatment may serve as an important tool for attenuating atrophy and preserving mitochondrial function in immobilized or injured athletes. Finally, the potential interplay between exercise, thermal interventions and epigenetics is discussed. Future studies must be encouraged to clarify how repeated thermal interventions (heat and cold) affect long-term exercise training adaptations and to determine the optimal modalities (i.e., method of application, temperature, duration, relative humidity, and timing).

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Factors associated with persisting post-concussion symptoms among collegiate athletes and military cadets: Findings from the NCAA-DoD CARE Consortium. Sports Medicine, Ahead of Print: 1-13, 2025.

Background: Persisting post-concussion symptoms (PPCS) is a condition characterized by prolonged recovery from a mild traumatic brain injury (mTBI) and compromised quality of life. Previous literature, on the basis of small sample sizes, concludes that there are several risk factors for the development of PPCS.

Objective: We seek to identify protective and risk factors for developing slow recovery or persisting post-concussion symptoms (PPCS) by analyzing medical history, contact sport level, setting, and the Sport Concussion Assessment Tool (SCAT) and Brief Symptom Inventory (BSI-18) assessments at baseline and post-injury.

Patients and methods: We studied 47,860 unique collegiate athletes and Military Service Academy (MSA) cadets enrolled in the 30-site National Collegiate Athletic Association and Department of Defense (NCAA-DoD) Concussion Assessment Research and Education (CARE) Consortium prospective cohort study of baseline and post-injury data from 2014 to 2019. Medical histories and preseason baselines (n = 60,720), along with SCAT and BSI-18 examinations (n = 5379) conducted within 48 h post-injury, were analyzed. PPCS is defined as cleared for return to play (RTP) > 29 days.

Results: Of 5073 concussions, the median (95% CI) and mean (SD) RTP were 15.2 (4.8-82.6) days and 22.2 (23.0) days, respectively, with 891 developing PPCS. Sex, high SCAT score, high BSI-18 score, and delayed reporting produced small effect sizes on RTP (d = 0.22-0.44). Adjusted odds ratios (OR) of developing PPCS indicated the following risk factors: SCAT total score > 45 (OR = 1.91, 95% CI: 1.58-2.30), female sex (OR = 1.80, 95% CI: 1.53-2.13), concussion history (OR = 1.80, 95% CI: 1.29-2.52), and delayed reporting (OR = 1.42, 95% CI: 1.20-1.67). In contrast, protective factors against developing PPCS were: being a limited-contact (OR = 0.34, 95% CI: 0.25-0.47), noncontact (OR = 0.35, 95% CI: 0.24-0.51) or contact sport athlete (OR = 0.34, 95% CI: 0.28-0.41); and receiving the concussion at practice (OR = 0.64, 95% CI: 0.53-0.77) or in competition (OR = 0.44, 95% CI: 0.34-0.56). Athletes diagnosed with a learning disability and taking a neurostimulant were more protected than those who were not (OR = 0.44, 95% CI: 0.26-0.76 versus OR = 0.72, 95% CI: 0.49-1.05, respectively). A prognostic model using these variables offers poor sensitivity (9%) but high specificity (98%) in identifying PPCS (AUC = 0.72).

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Conclusions: This study of 5073 concussions indicates that female sex, high symptom burden, and prior concussion are risk factors for slow recovery. In addition, learning disability, contact sport, and concussion in competition are protective against slow recovery. The neurostimulant results suggest that clinicians should keep their attention deficit disorder (ADD) and attention-deficit/hyperactivity disorder (ADHD) patients on their current neurostimulant medication after a concussion. A prognostic logistic regression model based on behavioral clinical findings did a poor job of identifying PPCS.

Key Points Findings:

Using 5073 concussed athletes, this case– control study finds that type of sport, injury setting, and learning disability are protective for persisting post-concussion symptoms, whereas sex, concussion history, high symptom burden, and delayed reporting are risk factors for PPCS.

Meaning: For the first time, this study provides evidence of factors that are protective against slow concussion recovery and suggests patients on neurostimulants should remain on their medication.

Football (soccer) match-derived hamstring muscles residual fatigue can be monitored using early rate of torque development. European Journal of Applied Physiology, Ahead of Print: 1-13, 2024.

Purpose: The aim of this study was to determine whether a soccer match affects the rapid force-generating capacity of the hamstring muscles, given their key role in both horizontal ground reaction force production during sprint biomechanics, and in the deceleration of the shank during the late swing phase, where rapid force production is essential owing to time constraints. Therefore, the research objective was to determine soccer match-induced hamstrings residual fatigue and recovery through rate of torque development (RTD) and associated biochemical parameters.

Methods: The recovery kinetics of hamstrings RTD metrics by the $90°_{hip}$:$20°_{knee}$ test, together with serum biomarkers (creatine kinase, mitochondrial creatine kinase, transaminases, malondialdehyde, irisin), were assessed in 19 male, regional first-division soccer players (age = 20.9 ± 2.0 years, mass = 72.6 ± 11.9 kg, height = 175.9 ± 6.9 cm [mean ± SD]), before a soccer match (MD) and post-24 h (MD_{+1}), post-48 h (MD_{+2}) and post-72 h (MD_{+3}), through a repeated measures design.

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Results: Early RTD to 50 ms ($p < 0.001$, $g = -1.24$) and 100 ms ($p < 0.001$, $g = -1.06$) remained unrecovered on MD+3 in both hamstring muscles. However, maximal voluntary isometric contraction (MVIC) torque of the dominant and non-dominant hamstrings was unrecovered on MD+2 ($p = 0.004$, $g = -0.91$; and $p = 0.002$, $g = -0.98$, respectively) and recovered on MD_{+3} ($p = 0.057$ and $p = 0.070$, respectively). Further, neuromuscular deficits were coupled with myocyte structural ($p = 0.002$, $g = 1.11$) and mitochondrial damage ($p = 0.004$, $g = 0.92$) biomarkers.

Conclusion: Based in the findings, early $RTD_{0\text{-}50}$ and $RTD_{0\text{-}100}$ monitoring, through the $90°_{hip}$:$20°_{knee}$ IPC test, is a cost-effective method for assessing soccer match-induced hamstring muscles residual fatigue and recovery. Overall, soccer match-induced hamstring residual fatigue is not recovered within a 3-day recovery period. Practitioners can use rapid force production metrics through isometric assessments, providing a simple, non-exhaustive tool, for assessing residual fatigue status during congested competitive periods, to comprehensively balance muscle recovery with optimizing training.

Fractional tackles: Leveraging player tracking data for within-play tackling evaluation in American football. Scientific Reports, 15: 2148, 2025.

Abstract - Tackling is a fundamental defensive move in American football, with the main purpose of stopping the forward motion of the ball-carrier. However, current tackling metrics are manually recorded outcomes that are inherently flawed due to their discrete and subjective nature. Using player tracking data, we present a novel framework for assessing tackling contribution in a continuous and objective manner. Our approach first identifies when a defender is in a "contact window" of the ball-carrier during a play, before assigning value to each window and the players involved. This enables us to devise a new metric called fractional tackles, which credits defenders for halting the ball-carrier's forward motion toward the end zone. We demonstrate that fractional tackles overcome the shortcomings of traditional metrics such as tackles and assists, by providing greater variation and measurable information for players lacking recorded statistics like defensive linemen. We view our contribution as a significant step forward in measuring defensive performance in American football and a clear demonstration of the capabilities of player tracking data.

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From food supplements to functional foods: Emerging perspectives on post-exercise recovery nutrition. Nutrients, 16: 4081, 2024.

Abstract - Effective post-exercise recovery is vital for optimizing athletic performance, focusing on muscle repair, glycogen replenishment, rehydration, and inflammation management. This review explores the evolving trend from traditional supplements, such as protein, carbohydrates, creatine, and branched-chain amino acids (BCAAs), toward functional foods rich in bioactive compounds. Evidence highlights the benefits of functional foods like tart cherry juice (anthocyanins), turmeric-seasoned foods, and sources of omega-3 fatty acids, including fish, flaxseeds, chia seeds, and walnuts, for mitigating oxidative stress and inflammation. Additionally, probiotics and prebiotics support gut health and immune function, which are integral to effective recovery. Personalized nutrition, informed by genetic and metabolic profiling, is examined as a promising approach to tailor recovery strategies. A systematic search across PubMed, Web of Science, and Google Scholar (2000–2024) identified studies with high empirical rigor and relevance to recovery outcomes. Findings underscore the need for further research into nutrient interactions, dosage optimization, and long-term effects on athletic performance. Integrating functional foods with personalized nutrition presents a comprehensive framework for enhanced recovery, greater resilience to physical stress, and sustained performance in athletes.

High rate of return to sport for athletes undergoing articular cartilage restoration procedures for the knee: A systematic review of contemporary studies. The American Journal of Sports Medicine, Ahead of Print: 1-12, 2025.

Background: Continued advancements in cartilage surgery and an accumulating body of evidence warrants a contemporary synthesis of return to sport (RTS) outcomes to provide updated prognostic data and to better understand treatment response.

Purpose: To perform an updated systematic review of RTS in athletes after knee cartilage restoration surgery.

Study design: Systematic review; Level of evidence, 4.

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Methods: PubMed, OVID/Medline, and Cochrane databases were queried in October 2023 for studies reporting any RTS outcome after knee cartilage restoration surgery. Meta-analyses with inverse-variance proportion and DerSimonian-Laird random-effects estimators were applied to quantify overall RTS. Comparative proportional subgroup meta-analyses with relative odds ratios (ORs) were constructed to quantify (1) the influence of the procedure on RTS and (2) RTS ability (lower vs same/greater level of play) based on procedure, competition level, and specific sport.

Results: A total of 52 studies (n = 2387) were included. The overall pooled RTS was 80.3% (95% CI, 73.3%-86.5%). Matrix-assisted chondrocyte implantation (MACI) (OR, 2.15) and osteochondral autograft transplantation system (OATS) (OR, 1.83) demonstrated the highest likelihoods of RTS at the same/greater level, while microfracture (MF) (OR, 0.78) was the only treatment demonstrating a higher likelihood of RTS at a lower level. The fastest mean RTS was observed after OATS (6.6 ± 2.6 months). Professional athletes demonstrated an OR of 1.01 for RTS at the same/greater level, whereas recreational/amateur athletes demonstrated an OR of 1.63; however, all professional athletes underwent MF, and recreational/amateur athletes who underwent MF demonstrated lower likelihoods of RTS (OR, 0.78), indicating a consistent association between MF and low RTS propensity. Basketball players demonstrated the lowest likelihood of RTS at the same/greater level (OR, 1.1), while American football and soccer were associated with high likelihoods of RTS (OR, 3 and 2.4, respectively) across all procedure types.

Conclusion: Cartilage restoration allows for high overall RTS, with OATS and MACI conferring the greatest propensity for RTS, while OATS allowed for the fastest RTS. Undergoing MF was associated with consistently poor RTS ability. This study identified several important associations between the level of RTS and clinically relevant factors when discussing RTS, with recreational/amateur athletes, soccer players, and American football players demonstrating a higher relative propensity to RTS.

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Histological definition of skeletal muscle injury: A guide to nomenclature along the connective tissue sheath/structure. Sports Medicine, Ahead of Print: 1-5, 2024.

Abstract - Recent years have seen the development of various classifications of muscle injuries, primarily based on the topographic location within the bone-tendon-muscle chain. This paper proposes an enhanced nomenclature for muscle injuries that incorporates histoarchitectural definitions alongside topographic classifications, emphasizing the importance of connective tissue damage characterization. A detailed understanding of the distinct anatomical and histological characteristics of tendon, aponeurosis, and fascia is essential for consistent terminology. Tendons, aponeuroses, and fasciae, while all composed of dense connective tissue, differ in collagen fiber orientation and structural organization. Tendons feature longitudinally aligned fibers suited for high tensile forces and muscle-to-bone connections. Aponeuroses have perpendicular collagen fiber layers, allowing for force distribution and support for both longitudinal and transverse traction. Fasciae exhibit loosely organized fibers providing flexibility and movement between muscles. The continuity and differences within the musculotendinous frameworks, particularly at the musculotendinous junction (MTJ), elucidate the higher frequency of injuries at these sites compared with tendons, highlighting the MTJ's role as a stress-absorbing 'safety valve.'

Key Points:

Dense connective tissue structures are highly implicated in muscle injuries in sports.

There is still a lack of knowledge of the appropriate usage of common anatomical terms like tendon, aponeurosis, or fascia.

By focusing on the histological characteristics, there is a layer of specificity added that can significantly refine the approach to diagnosis, prognosis, and treatment of sports-related muscle and tendon injuries

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Knee ligament injuries in the National Football League: Impact of field surface type during the 2020-2-2023 season. Cureus, 16(12): e76425, 2024.

Introduction: A subject of ongoing debate within the National Football League (NFL) community revolves around the comparative risk of anterior cruciate ligament (ACL) injuries on natural versus artificial turf field surfaces. There have been mixed results as to whether there is a difference in injury rates depending on the playing surface and what factors might play a role in affecting these rates.

Methods: This study aims to compare the incidence of in-game knee ligament tears in the NFL during the 2020-2023 seasons. Data was collected from publicly available resources to determine and classify specific injury characteristics such as ligament, injury date, location, and field surface type. Only in-game knee ligament tears, including those occurring in pre-season games, were included. Significance was determined using two-sample T-tests.

Results: Over the 2020, 2021, 2022, and 2023 seasons, there were 149 ligament tears, with 58 occurring on natural turf (38.9%) and 91 occurring on artificial turf (61.1%). Over the course of the same four years, 1213 games were played, with 524 (43.2%) games on natural turf and 689 games on artificial turf (56.8%). Accounting for the number of games played on each surface, natural turf saw a rate of 0.111 knee ligament tears per game, while artificial turf saw 0.132 ligament tears per game.

Conclusion: Over the course of the study period, we determined that there was no significant difference between the rates of in-game knee ligament tears on natural and artificial turf playing surfaces. Our study shows that regardless of differences in composition and other playing surface features, there is ultimately no difference in the incidence of knee ligament tears. Playing surfaces should still be monitored and regulated, and the incidence of other injuries, such as concussions, as well as player preference, should be considered when considering the ideal surface.

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Malpractice liability exposure and the sports medicine team physician: Caring for professional athletes in the National Football League, Major League Baseball, and National Hockey League. The American Journal of Sports Medicine, Ahead of Print: 1-8, 2025.

Background: Orthopaedic surgeons play a critical role in ensuring the health and safety of professional athletes. Despite the privilege of treating elite athletes, there exists great financial exposure to individual physicians in the event of a malpractice lawsuit.

Hypothesis/purpose: The purpose of this study was to evaluate and model malpractice liability exposure of the sports medicine surgeon caring for athletes in the National Football League (NFL), Major League Baseball (MLB), and National Hockey League (NHL) with respect to player position and additional supplemental malpractice insurance needs. It was hypothesized that routine liability coverage cannot adequately address the demands of caring for elite athletes in professional sports leagues.

Study design: Economic and Decision Analysis; Level of evidence, 3.

Methods: In total, 2447 NFL, 992 MLB, and 980 NHL player contracts from the 2022-2023 season were aggregated from a publicly available online database. Position, team, total contract value, and mean yearly salary were noted. Risk ratios were calculated with respect to 1 million US dollars (USD) and 3 million USD of annual occurrence-based malpractice liability awards and used to generate a "covered-to-treat" analysis. Supplemental malpractice liability insurance was quantified.

Results: Assuming 1 million and 3 million USD occurrence-based awards covered by malpractice liability insurance, team physicians can fully treat 17.3% and 50.0% of NFL players, 43.2% and 59.7% of MLB players, and 13.6% and 41.0% of NHL players, without incurring additional personal financial risk, from a risk-based medicolegal model. Liability policies of 52.6 million, 108.1 million, and 64.1 million USD are required to treat 95% of NFL, MLB, and NHL players, respectively. Positions carrying the greatest risk ratios are quarterback (QB) (9.9) in the NFL, right field (15.1) in the MLB, and center (5.7) in the NHL.

Conclusion: Sports medicine specialists caring for elite athletes face potential personal financial risk due to insufficient medicolegal coverage. While coverage may vary among different practice settings including private, academic, or public state institutions, medical malpractice risk is crucial in

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partnerships between sports franchises, hospitals, players, agents, and physicians to protect sports medicine physicians and offer the highest-quality care.

Nutrition for optimizing immune function and recovery from injury in sports. Clinical Nutrition ESPEN, Ahead of Print: 1-37, 2025.

Abstract - The immune system, sports injuries, physical activity and clinical sports nutrition are closely linked. Inadequate nutrition and intense stress in athletes increase the risk of immune dysfunction, injury and disease, while injury reduces the positive effects of physical activity, creating a vicious cycle affecting health and performance. Nutritional support is key to reducing the risk of injury, speeding up recovery and shortening rehabilitation. The intake of energy, protein, carbohydrates, fats and micronutrients should be adapted to the needs following injury. Quality nutrition is preferred over the use of dietary supplements, which are limited to cases of identified deficiencies (e.g. vitamin D or iron deficiencies). Health professionals with knowledge of clinical nutrition have a key role to play in injury prevention, treatment and rehabilitation, and nutritional interventions are an indispensable part of a holistic approach to athletes' health.

Orthopedic patients are highly amenable to cannabis-based products for the treatment of musculoskeletal pain. Orthopedics, Ahead of Print: 1-8, 2025.

Background: Tetrahydrocannabinol (THC)-based and cannabidiol (CBD)-based products are increasing in popularity for the management of pain. Cannabis-based products may serve as a valuable alternative to traditional analgesics such as opioids for pain management among orthopedic patients. The purpose of this study was to investigate the perspective of orthopedic sports medicine patients regarding the use of THC- and CBD-based products for the management of musculoskeletal pain.

Materials and methods: A short, 11-question survey was distributed to orthopedic sports medicine patients in clinic and via email. Survey participation was voluntary. Retrospective chart review was conducted for participating patients to collect additional data such as visit diagnosis.

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Results: A total of 182 patients completed the survey. Most were familiar or somewhat familiar with THC (61%) and CBD (63.2%) products and their use for musculoskeletal pain. Of the patients, 53.3% were aware of friends and family currently using CBD products to help manage musculoskeletal pain. Of the patients, 81.3% were amenable to receiving THC-based products and 90.1% were amenable to receiving CBD-based products for the management of their musculoskeletal pain. The patients were statistically more amenable to receiving CBD-based products than THC-based products (P=.017). Among the patients, 85.3% believed cannabis-based products would help in the fight against the opioid epidemic.

Conclusion: Orthopedic patients are aware of THC- and CBD-based products, and the majority are willing to receive these products to help manage their musculoskeletal pain and orthopedic-related issues. Orthopedic surgeons should be aware of the increasing popularity of CBD-based products and educate themselves and their patients regarding the appropriate use of them.

Performance effects of functional knee brace removal and prolonged use in healthy male athlete: Lower extremity power, acceleration, speed, and agility. Journal of Science and Medicine in Sport, Ahead of Print: 1-6, 2025.

Objectives: Knee braces were introduced to sports 30 years ago. However, knee brace use for non-contact anterior cruciate ligament injury prevention intervention remains contentious due to concerns about performance hindrances. Since knee brace use is a potential modifiable risk factor, we aimed to investigate the effect of discounting and continued functional knee brace (FKB) on lower extremity power-vertical jump (VJ), acceleration, speed, and agility performance.

Design: Prospective cohort crossover study.

Methods: Twenty-seven healthy male athletes performed seven tests, over six days of 12 test sessions (S), during three test conditions (non-braced, braced, and removed brace or continued brace use). This study focuses on VJ, acceleration, speed, and agility performance during S12 when athletes were randomly selected to remove the FKB after 17.5 h or continue using the FKB for 21.0 h.

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Results: After brace removal, nonsignificant performance levels improved in the VJ (2.7 %; 95 % CI 52.5-62.8; Cohen's effect size (ES) = trivial), acceleration (1.8 %; 95 % CI 0.500-0.562; ES = small), and agility (0.5 %; 95 % CI 9.25-10.13; ES = trivial), while a nonsignificant slower speed was recorded (0.5 %; 95 % CI 1.81-1.95; ES = trivial). Continued brace use led to a nonsignificant performance improvement in all tests; VJ (3.1 %; 95 % CI 53.5-60.2; ES = small), acceleration (1.5 %; 95 % CI 0.511-0.561; ES = trivial), speed (1.0 %; 95 % CI 1.83-1.95; ES = trivial), and agility (1.8 %; 95 % CI 9.26-10.04; ES = trivial).

Conclusions: Removal of FKB led to improved performance in three performance tests, while continued brace use improved performance in all four tests.

Predicting noncontact injuries of professional football players using machine learning. PLOS ONE, 20(1): e0315481, 2025.

Abstract - Noncontact injuries are prevalent among professional football players. Yet, most research on this topic is retrospective, focusing solely on statistical correlations between Global Positioning System (GPS) metrics and injury occurrence, overlooking the multifactorial nature of injuries. This study introduces an automated injury identification and prediction approach using machine learning, leveraging GPS data and player-specific parameters. A sample of 34 male professional players from a Portuguese first-division team was analyzed, combining GPS data from Catapult receivers with descriptive variables for machine learning models-Support Vector Machines (SVMs), Feedforward Neural Networks (FNNs), and Adaptive Boosting (AdaBoost)-to predict injuries. These models, particularly the SVMs with cost-sensitive learning, showed high accuracy in detecting injury events, achieving a sensitivity of 71.43%, specificity of 74.19%, and overall accuracy of 74.22%. Key predictive factors included the player's position, session type, player load, velocity and acceleration. The developed models are notable for their balanced sensitivity and specificity, efficiency without extensive manual data collection, and capability to predict injuries for short time frames. These advancements will aid coaching staff in identifying high-risk players, optimizing team performance, and reducing rehabilitation costs.

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Pseudo-sciatica pain due to a biceps femoris muscle tear. Cureus, 16(11): e74286, 2024.

Abstract - Hamstring injuries (HSIs) are common among athletes, particularly in high-speed sports, and are frequently associated with prolonged recovery periods and a high recurrence rate. This study presents a 36-year-old male with sciatica-like symptoms following an acute hamstring tear during an amateur football match. The patient experienced this pain two weeks after the injury, suggesting potential sciatic nerve involvement. Ultrasonographic imaging revealed a partial myotendinous rupture of the biceps femoris long head, adjacent to the sciatic nerve. The patient underwent a structured five-week rehabilitation program, resulting in significant improvement, including reduced pain, enhanced muscle strength, and improved flexibility. This case highlights the role of early intervention and targeted rehabilitation in preventing complications, such as hamstring syndrome (HSS), and emphasizes the importance of using ultrasonography for diagnosis. This study also addresses the limitations in diagnostic imaging and provides insights into the link between muscle injury and sciatic nerve irritation, contributing valuable knowledge to the limited literature on this subject.

Relationship between serum Vitamin D levels and chronic musculoskeletal pain in adults: A systematic review. Nutrients, 16: 4061, 2024.

Abstract - Background/Objectives: Chronic pain impacts approximately 18% of the Spanish population, with low levels of vitamin D prevalent in over 80% of individuals over 65. Given vitamin D's critical role in pain modulation, its deficiency may be significantly linked to chronic musculoskeletal pain, though existing research offers mixed results. Methods: This systematic review followed PRISMA guidelines, examining studies from PubMed, Cochrane, and PEDRO databases from 1990 onwards that investigated the relationship between vitamin D levels and chronic musculoskeletal pain. Results: A total of 30 studies met the inclusion criteria set by the NHLBI's quality standards. The results are inconclusive regarding the direct relationship between vitamin D levels and chronic musculoskeletal pain due to evidence heterogeneity. However, there appears to be an inverse relationship between vitamin D levels and the intensity of pain. Conclusions: While the association between vitamin D levels and chronic musculoskeletal pain remains uncertain, the inverse correlation with pain intensity suggests a potential

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therapeutic role of vitamin D supplementation in pain management. Further research is needed to substantiate these findings and refine intervention strategies.

Return to sport after distal biceps tendon repair: A systematic review. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2025.

Background: Ruptures of the distal biceps tendon (DBT) can affect the range of motion and strength of the elbow, raising concerns for patients seeking to restore normal function and engage in their regular activities, particularly returning to previous levels of sport participation.

Purpose: To characterize and assess the rate and timing of return to sport (RTS) after DBT repair.

Study design: Systematic review; Level of evidence, 4.

Methods: PubMed, Embase, and Google Scholar (pages 1-20) were searched from database inception to December 6, 2023 for clinical studies reporting RTS outcomes after DBT repair. The extracted data consisted of patient characteristics; information on the incision approach, fixation method, and rehabilitation protocol; and outcome data including RTS rates, patient-reported outcome measure scores, and complications.

Results: A total of 42 studies, including 1093 patients (1100 elbows), met the inclusion criteria. The mean age of patients was 44.9 years, and 99.2% of patients were male. The mean follow-up time was 31.5 months. Ruptures were acute in 87.6% of cases, the dominant arm was injured in 64.9%, and the cause of the injury was sport related in 43.5%. The overall RTS rate was 91.5%, with 85.2% of patients returning to preinjury levels or higher, at a mean time of 6.3 months. Patients had excellent functional outcomes, irrespective of the incision approach or fixation method, although trends associated with a higher RTS rate were observed with bone tunnel fixation, ≤2 weeks of postoperative immobilization, early initiation of active range of motion postoperatively, and initiation of strengthening at ≤10 weeks. Single-incision repair had higher rates of nerve-related complications and reruptures compared with double-incision repair, and cortical button fixation had a higher rate of nerve-related complications among the fixation methods.

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Conclusion: There was a high rate of RTS after DBT repair at 6 months postoperatively. A positive trend for RTS was observed with respect to rehabilitation protocols favoring earlier active mobility.

Role of hip internal rotation range and foot progression angle for preventing Jones fracture during crossover cutting. Journal of Applied Biomechanics, Ahead of Print: 1-7, 2024.

Abstract - Jones fracture is a common injury in soccer players, caused by overload on the fifth metatarsal (M5) that can occur during crossover cutting. This study aimed to investigate the effects of the hip internal rotation (HIR) range in passive and dynamic conditions and foot progression angle (FPA) on the forces applied to the M5 during crossover cutting. The study included 20 men with experience playing soccer. The passive HIR was measured in the prone position. A 3-dimensional motion analysis system, force plates, and Footscan were used to measure the angle of the lower limb, including dynamic HIR, FPA, and plantar pressure to the M5, during crossover cutting in the 45° direction. Pearson correlation coefficient was measured to examine the relationship between variables. Passive and dynamic HIR were not related to the plantar pressure on the M5 ($P > .05$). However, increased FPA in movement direction correlated with a reduction in plantar pressure on the M5 ($r = -.56$, $P < .01$). Minimizing the FPA with respect to the new direction during crossover cutting reduced plantar pressure on the M5 and thus could prevent Jones fracture.

Sleep disturbances and risk of sports injury among collegiate student-athletes. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

Objective: The purpose of this study was to examine the relationship between collegiate student-athletes' sleep and injury status.

Design: This study was a secondary analysis of observational survey data of student-athletes collected across 2 time periods, October 2020 and January 2021.

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Setting: Data were collected as part of an ongoing national (USA) longitudinal survey investigating student-athletes' well-being in the aftermath of the COVID-19 pandemic.

Participants: Participants were made-up of a subsample of participants from the parent study. Data from 89 exact matched pairs (across 5 demographics) of injured and noninjured student-athletes (68.5% NCAA Division I, 82.0% women) were assessed.

Independent variables: Student-athletes' sleep disturbance levels were the primary independent variable of interest. Analytic models also covaried for gender, racial, and ethnic identities, sport in-seasons, and previous injury status.

Main outcome measures: Participant endorsed injury status was the main outcome measure of interest. Hypotheses were formulated during the data collection of the parent study.

Results: Student-athletes injured in January 2021 reported higher average sleep disturbances in both October 2020 and January 2021 than noninjured student-athletes but not student-athletes injured in October. In addition, October 2020 sleep disturbances were a significant predictor ($P = 0.007$) of injury, such that odds of student-athletes being injured increased 1.07 times for each unit increase in their sleep disturbances controlling for their identities, sport in-season, and previous injury status.

Conclusions: Findings signal support for the screening and targeting of student-athletes' sleep problems to reduce potential occurrences of sports-related injuries in student-athletes.

Sports Medicine: Nutritional sciences and nutritional biochemistry, focusing on nutritional supplements from the FFNFO Framework to contamination. Life, 14: 1392, 2024. (No abstract available).

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The effect of alcohol on subsequent sleep in healthy adults: A systematic review and meta-analysis. Sleep Medicine Reviews, 80: 102030, 2025.

Abstract - Alcohol is commonly consumed prior to bedtime with the belief that it facilitates sleep. This systematic review and meta-analysis investigated the impact of alcohol on the characteristics of night-time sleep, with the intent to identify the influence of the dose and timing of alcohol intake. A systematic search of the literature identified 27 studies for inclusion in the analysis. Changes in sleep architecture were observed, including a delay in the onset of rapid eye movement (REM) sleep and a reduction in the duration of REM sleep. A dose-response relationship was identified such that disruptions to REM sleep occurred following consumption of a low dose of alcohol (≤0.50 g·kg-1 or approximately two standard drinks) and progressively worsened with increasing doses of alcohol. Reductions in sleep onset latency and latency to deep sleep (i.e., non-rapid eye movement stage three (N3)) were only observed following the consumption of a high dose of alcohol (≥0.85·g kg-1 or approximately five standard drinks). The effect of alcohol on the remaining characteristics of sleep could not be determined, with large uncertainty observed in the effect on total sleep time, sleep efficiency, and wake after sleep onset. The results of the present study suggest that a low dose of alcohol will negatively impact (i.e., reduce) REM sleep. It appears that high doses of alcohol may shorten sleep onset latency, however this likely exacerbates subsequent REM sleep disruption. Future work on personal and environmental factors that affect alcohol metabolism, and any differential effects of alcohol due to sex is encouraged.

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The significance of body surface area to mass ratio for thermal responses to a standardized exercise-heat stress test. Medicine and Science in Sports and Exercise, 57(1): 88-93, 2025.

Purpose: To evaluate the significance of body surface area-to-mass ratio (BSA/mass) on the heat-tolerance test (HTT) results. We hypothesized that individuals defined as heat tolerant (HT) would have on average higher BSA/mass compared with heat intolerant (HI) individuals.

Methods: A retrospective reanalysis of the HTT results of 517 soldiers (age, 18-38 yr; M/F, 96%:4%), who were tested by the Israel Defense Forces HTT protocol. The criterion for heat tolerance in the current analysis was a rectal temperature (T re) plateau during the second hour of the test. A logistic regression analysis to evaluate the predictive power of BSA/mass for heat intolerance was performed; the spline model was applied to show the odds for heat intolerance across BSA/mass.

Results: In men BSA/mass of HI individuals was lower than HT individuals (248 ± 19 vs 262 ± 18 cm 2 ·kg -1 , $P < 0.01$, $d = 0.76$). In women a similar trend was noted but with no statistical significance between HT and HI groups. The odd ratio for heat intolerance for every unit increase in BSA/mass was 0.97 (95% confidence interval, 0.95-0.99). The spline model plateaued above BSA/mass of 270 cm 2 ·kg -1 .

Conclusions: The results imply that body-core temperature responses to a standard exercise-heat stress (fixed external work rate and climatic conditions) are influenced by BSA/mass. More specifically, lack of a steady state in T re (indicating heat intolerance) was more likely to occur with every unit decrease in BSA/mass. These findings contribute to a better understanding of the role of body anthropometry in the response to a standard exercise-heat task that might have an implication on clinical decision making about return to duty/play of soldiers, athletes, and others who deemed to be identified as HI.

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The value of blood lactate and lactate clearance rate in evaluating the prognosis of athletes with heat illness of varying degrees after high-intensity exercise. BMC Sports Science, Medicine and Rehabilitation, 17: 7, 2025.

Background: Heat stroke, a severe heat illness with organ damage, is a major cause of cause irreparable organ damage and higher death rates among military persons and athletes.

Objectives: To study the changes in blood lactate (Lac) levels and lactate clearance rate (LCR) in athletes with heat illness of varying degrees after high-intensity exercise and to evaluate their prognostic value.

Material and methods: In present study, acute care unit admitted 36 heat sickness patients following high-intensity exercise from December 2019 to July 2024, with comprehensive medical records, for retrospective study. The study population consisted of two groups of high level athletes: the favourable Prognosis Group (< 7 days, 22 cases), comprising 21 males and 1 female with a mean age of 21.8 ± 2.7 years, and the bad Prognosis Group (≥ 7 days, 14cases), consisting of 14 males with a mean age of 22.6 ± 3.2 years. Lac levels were assessed at admission (0 h) and early in therapy (2 h, 6 h), and the LCR was computed. Lac and LCR values at each time point were compared between the two groups to see how they affected patient prognosis.

Results: After 2 and 6 h of therapy, lactate levels decreased significantly in the good prognosis group (1.2 ± 0.5 mmol/L at 2 h and 0.8 ± 0.3 mmol/L at 6 h), but remained elevated in the poor prognosis group (4.2 ± 1.2 mmol/L at 2 h and 3.5 ± 1.5 mmol/L at 6 h). Core body temperature normalized in both groups, but the good prognosis group showed a more rapid decline, with temperatures of 37.4 ± 0.6 °C at 2 h and 36.8 ± 0.4 °C at 6 h in the good prognosis group, and 38.8 ± 0.8 °C at 2 h and 38.2 ± 0.9 °C at 6 h in the poor prognosis group. Notably, a significant positive correlation existed between lactate levels and APACHE II scores at admission ($P < 0.01$). Furthermore, logistic regression analysis revealed that the 2-hour lactate clearance rate (LCR) ($R2 = 0.83$) was an independent predictor of outcomes.

Conclusions: The study suggests that athletes with elevated lactate levels after heat illness may be at higher risk of adverse outcomes. The 2-hour lactate clearance rate (LCR) appears to be a valuable prognostic indicator, with potential applications in evaluating the severity of heat illness and guiding treatment decisions. Furthermore, dynamic monitoring of lactate levels in conjunction with LCR may

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provide valuable insights into the clinical management and prognosis of athletes with heat-related illnesses.

Ultrasound as a predictor of time-loss injury for the patellar tendon, Achilles tendon and plantar fascia in Division 1 collegiate athletes. British Journal of Sports Medicine, 59: 1-8, 2025.

Objectives: Tendinopathy and fasciopathy are common conditions that can result in time-loss injury in athletes. This study aimed to determine if preseason sonographic abnormalities of the patellar tendon, Achilles tendon and plantar fascia are associated with future time-loss injuries in collegiate athletes.

Methods: National Collegiate Athletic Association Division I athletes from three institutions participated in this 3-year prospective, observational study. Each athlete completed a questionnaire, self-reporting current and prior symptoms and underwent an ultrasound examination of bilateral patellar tendons, Achilles tendons and plantar fasciae during annual preseason examinations. Ultrasound evaluations assessed for tendon and fascia thickening, hypoechogenicity and neovascularisation. Athletes were monitored for time-loss injury over the subsequent year.

Results: A total of 695 athletes across 18 sporting disciplines (61.1% female, age 20.0±1.6, body mass index 23.1±2.9 kg/m2) were analysed over 3 years. Sonographic abnormalities were identified in 36.6%, 7.5% and 2.8% of the patellar tendons, Achilles tendons and plantar fasciae, respectively. Injuries were reported in 3.3%, 1.6% and 0.7% of these structures with an adjusted relative risk of injury increased by 8.9 (95% CI 3.7, 21.4), 18.8 (95% CI 7.2, 48.8) and 21.0 (95% CI 6.4, 68.1) times in those with preseason ultrasound abnormalities ($p<0.001$). The presence of an ultrasound abnormality was more predictive of future injury than self-report of a prior injury or pain in the area at the time of the scan.

Conclusion: Preseason sonographic abnormalities of the patellar tendon, Achilles tendon or plantar fascia are associated with a higher risk of developing time-loss injuries in collegiate athletes.

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Pro Football Sports Science Research Update

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Achilles and patellar tendons with sonographic abnormalities are more likely to progress to pain and injury than structurally normal tendons. A history of prior injury is also associated with future injury risk for these structures.

WHAT THIS STUDY ADDS

⇒ This prospective study of collegiate athletes demonstrates that sonographic abnormalities of the Achilles tendon, patellar tendon or plantar fascia are independently associated with an increased risk of time-loss injury. The risk associated with sonographic abnormalities appears to be higher than that with self-reported pain or prior injury at the time of scanning. Though the negative predictive value of a normal ultrasound examination is high, the positive predictive value of an abnormal ultrasound examination is low, suggesting that not all athletes with sonographic abnormalities will actually become injured.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ A preseason ultrasound evaluation of the Achilles tendon, patellar tendon and plantar fascia may identify at-risk athletes, particularly in the collegiate athlete population. A normal ultrasound is highly correlated to staying injury-free for that particular structure during the following year.

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